Ref 777

Palaeomagnetic data and radiometric ages from the Cretaceous of West Kalimantan (Borneo), and their significance in interpreting regional structure

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SUMMARY

Potassium-argon ages of biotite and hornblende from sixteen samples of mainly granitic plutonic rocks from West Kalimantan indicate a complex history of magmatism and cooling from the Middle Jurassic to late Cretaceous, with most ages in the range 75 to 112 Ma. Samples from the Schwaner Zone in the northern part of the area give measured ages ranging from 76 to 116 Ma. In the southern region most samples are in the range 79 to 86 Ma, but two are older (127 Ma, early Cretaceous, and 153 Ma, Middle Jurassic). In both areas, concordant ages from biotite and hornblende indicate a well-defined event at about 80 to 85 Ma, which has also been recognized to the north in Sarawak, on islands of the South China Sea, in the Malay Peninsula, south Sumatra, and in the Java Sea.

Palaeomagnetic measurements on thirty-nine samples of dyke rocks, lavas, tuffs, granites, hornfels and metatuff, all believed to be late Cretaceous, yield a mean palaeomagnetic pole at 21°E 41°N, and a palaeolatitude for West Kalimantan of 0°. The palaeomagnetic pole does not differ significantly from the Cretaceous pole estimated for the Malay Peninsula. It is concluded that since the Middle Cretaceous West Kalimantan and the Malay Peninsula have behaved as a unit, have remained in much their present latitude, but have rotated anticlockwise about 50°.

BORNEO LIES on the continental side of the Sunda Island arc of Sumatra and Java in a position difficult to reconcile with any simple geotectonic pattern, either in terms of geosynclinal or plate tectonic theory. The western part of the island, south of the Kapuas River, is formed of the largest plutonic complex in the Indonesian Archipelago, with associated hypabyssal, volcanic and metamorphic rocks, and sedimentary rocks of small extent. This igneous complex is flanked to the northwest and southeast by mobile belts of late Cretaceous to Cenozoic age, and has been called the "West Borneo Basement" (Haile 1973). The complete absence of geochronologic or palaeomagnetic evidence from this important west and central part of Borneo has hampered understanding of the geological history of the island and of the geotectonics of SE Asia as a whole. In recent years a number of attempts have been made to reconstruct the history of western Indonesia in terms of plate tectonic theory, and the present investigation was made in the hope of providing factual constraint to test these. The present investigation is confined to the part of the "West Borneo basement" west of 111°E and south of the equator (Fig. 1).

Rock samples collected in the present investigation are housed in the museum of the Department of Geology, University of Malaya, with duplicates of orientated specimens and of most others in the Research School of Earth Sciences, Canberra.

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1. Geology

THE ONLY SYSTEMATIC geological survey done in this area was by L. H. Krol and collaborators between 1925–1929. Krol died before he could publish his results, and the report and map by van Bemmelen (1939) is based on examination of more than 2500 rock specimens collected by Krol.

Van Bemmelen divided the region south of the Kapuas into two zones (Fig. 2): The Schwaner Range Zone, or Schwaner Zone, mostly occupied by tonalitic plutonic rock, and the "southern region" composed of flysch, contact-metamorphosed limestone, and volcanic rocks.

A. THE SCHWANER ZONE

The Schwaner Zone extends east from Pontianak, and is mostly occupied by batholiths of quartz diorite, tonalite and granite, referred to here in general as tonalite. Also present are gabbro, diorite, and norite. Dykes of lamprophyre, aplite, and pegmatite are common but (van Bemmelen notes) dolerite dykes are not seen there, although these occur in the zones to the north and south. Fresh basaltic, andesitic, dacitic, and rhyolitic dykes occur, and may be early Tertiary. Van Bemmelen recorded a few occurrences of dynamo-metamorphic quartzite and phyllites, possibly Upper Triassic, some hornfels, which he considered to be probably Permo-Carboniferous in age, and some "Crystalline Schist."

In the present investigation, the rocks K77–78 (Fig. 3) and all rocks collected to the north of this appear to belong to the Schwaner Zone. The only rock suitable for palaeomagnetic work found there is the basaltic dyke (K50 and 51) from the Keriau headwaters. All the plutonic rocks (except the hornblende gabbro) contain biotite, and most have hornblende. Enough transitional types occur to suggest that all the plutonic rocks are co-magmatic.

B. THE SOUTHERN REGION

Plutonic rocks are extensive but less dominant than in the Schwaner Zone. Van Bemmelen distinguished the following units:

Ketapang Complex. Flysch-like alternation of sandy shales and quartzitic sandstone (possibly Triassic on correlation with the Upper Triassic to the north and in Malaya). Contact metamorphic limestone, "possibly Permo-Carboniferous" (not found in the area discussed here).

Matan Complex. Volcanic rocks, mostly strongly folded. Probably in part the same age as the Ketapang Complex, but may be in part younger (even including Tertiary?). Mostly older than the granite batholiths.

Crystalline Schists. Doubtful occurrences only. Plutonic rock samples in the present investigation include alkali granites, with biotite (K23, K24, K29) some containing hornblende in addition (K16, K19, K46, K56), and granite porphyries K20, K21, K22). Adamellite and granodiorite from Pulau Cempedak (K1, K8, K14B) contain biotite and pyroxene, and are there associated with dykes of porphyritic quartz microdiorite (K6, K11).

Basic dyke rocks in the southern region are microgabbro which is mostly

olivine bearing (K2, K3, K13, K25, K26, K27, K28, K63, K64); basalt (K50, K51, K61, K62); and andesite (K53, K54, K55).

2. Geochronology

Sixteen of the least altered samples that contained sufficient biotite and/or hornblende, and covered as wide an area as possible, were selected for K-Ar age measurements. The analysed samples are plutonic rocks, mainly granitic, as all the volcanic and dyke rocks were found to be too altered. The results are given in Table 1 and plotted on Fig. 3. Potassium was determined by flame photometry and argon by isotope dilution using a substantially modified AEI MS10 mass spectrometer with a digital readout system for the measurement of isotopic ratios.

The physical time scale employed is that of Lambert (1971) who suggested the following ages for the relevant boundaries: Triassic-Jurassic, 200 Ma; Jurassic-Cretaceous, 135 Ma; early Cretaceous—late Cretaceous, 95 Ma; Cretaceous—Tertiary, 65 Ma.

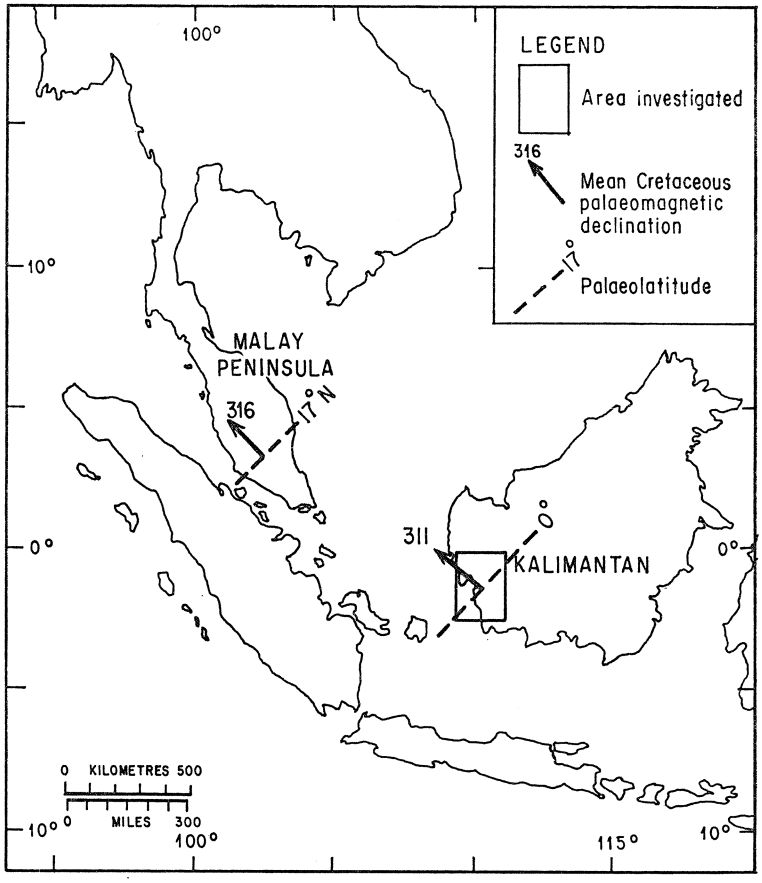


FIG. 1. Location map, mean Cretaceous palaeomagnetic declination, and palaeolatitude of West Kalimantan and the Malay Peninsula.

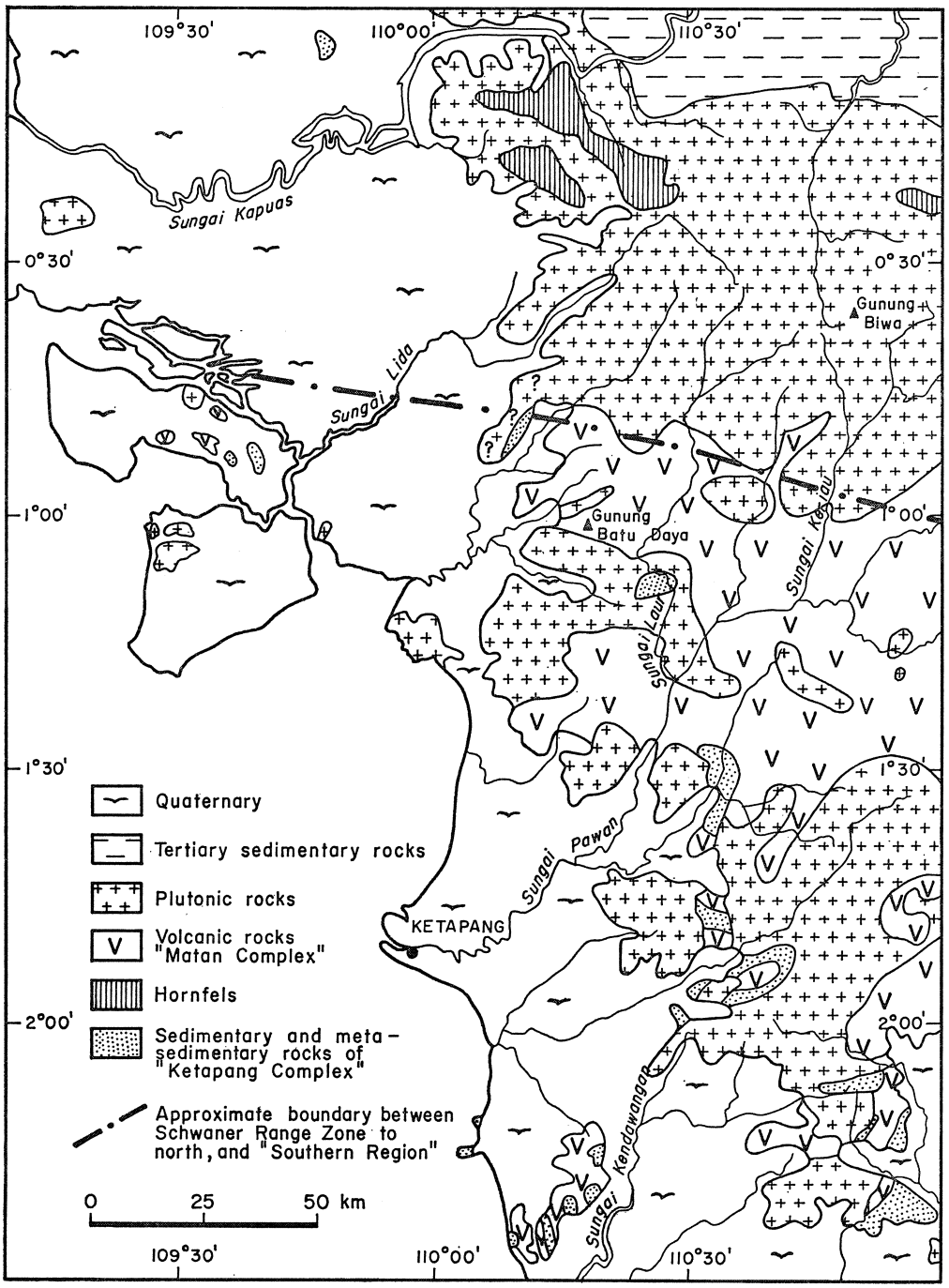


FIG. 2. Geological sketch map of part of West Kalimantan.

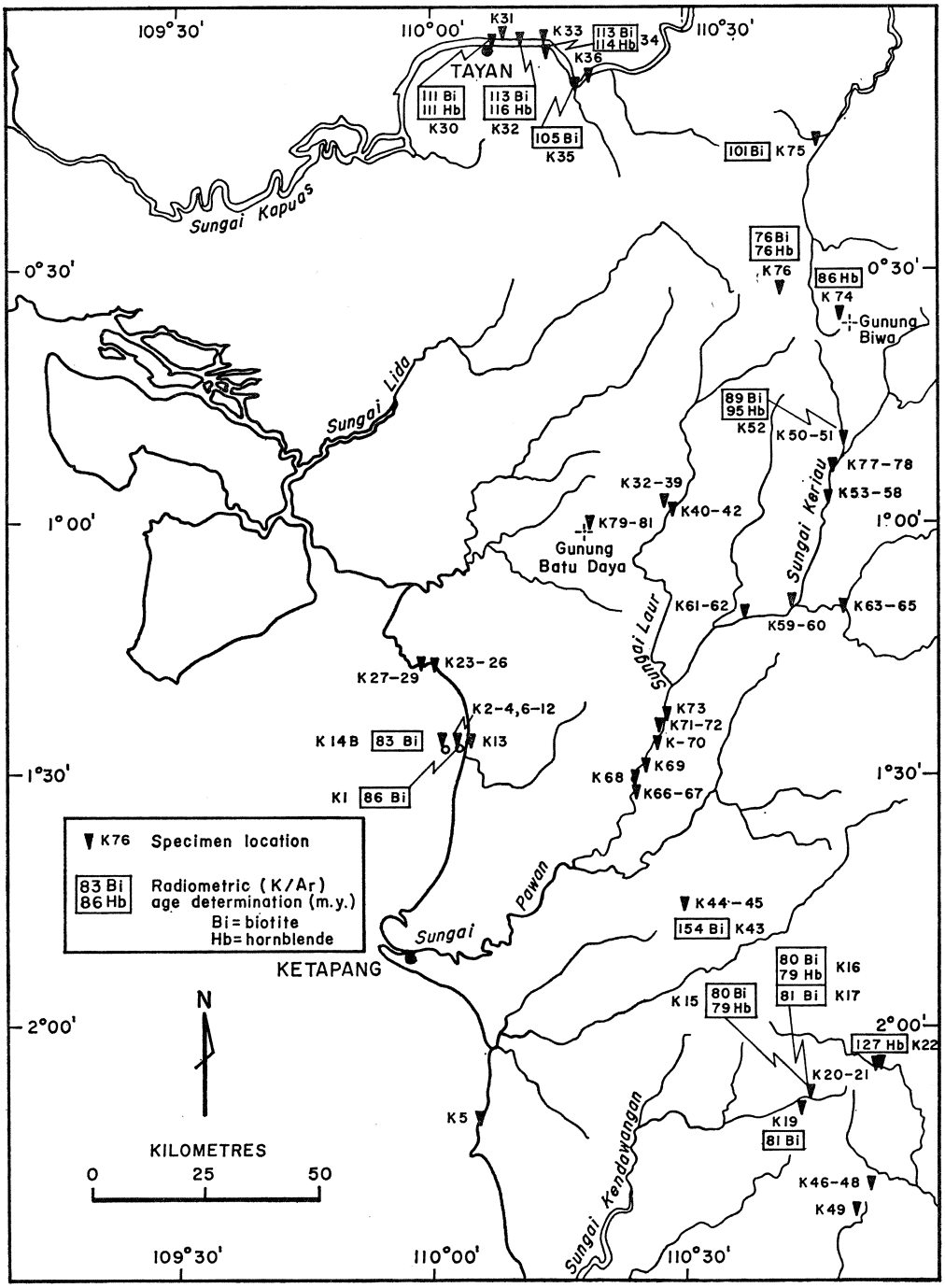


FIG. 3. Specimen location maps and radiometric ages, West Kalimantan.

A. RESULTS AND INTERPRETATION

Schwaner Zone. The zone north of 1° south latitude in Fig. 2 with Gunung Biwa near the eastern margin consists predominantly of plutonic, mainly granitic rocks which crop out over a roughly circular region about 100 km in diameter. There is little information as to the number of intrusive bodies or their relation to one another. A total of eight samples were used in the K-Ar dating of this large region, so that the results at best are only likely to give a rough indication as to the geological age of the rocks.

The measured ages on biotite and hornblende show a wide scatter from 76 Ma to 116 Ma (early to late Cretaceous). In those cases where hornblende and biotite were measured from the same rock the agreement between the ages generally is good, providing confidence that real geological events are being recorded. The major question is whether these ages record the time of emplacement and crystallization of these plutonic rocks, or some subsequent event such as rapid cooling consequent upon uplift from deeper parts of the earth's crust. Such uplift could occur much later than the time of emplacement. Empirical observations from contact metamorphic zones suggest that hornblende is more retentive of radiogenic argon than biotite (Hart 1964, Hanson & Gast 1967). From this it may be inferred that hornblende is likely to begin to retain radiogenic argon quantitatively at a considerably higher temperature than biotite during cooling of an igneous body. Thus the fact that the hornblende and biotite ages agree well in most cases therefore can be interpreted to mean that the cooling event is likely to have been rapid, but in the absence of much additional data, which ideally should include Rb-Sr results, it is not possible to choose unequivocally between emplacement and uplift ages. There is no evidence from examination of thin sections of these rocks for events subsequent to crystallization except for minor to moderate alteration of feldspars and variable chloritization of biotite, all of which may be deuteric.

Nevertheless, the present reconnaissance geochronological study suggests that significant geological events related to emplacement of plutonic bodies and their ultimate cooling took place over an age range from about 113 Ma to 76 Ma ago i.e. in the later part of the early Cretaceous into the late Cretaceous. The country rocks into which the plutonics are emplaced therefore are early Cretaceous or older.

The Southern Region. This region is as large as and perhaps even more diverse geologically than the Schwaner Zone to the north. The K-Ar ages on hornblende and biotite show an even greater spread from about 80 Ma (late Cretaceous) to to 154 Ma (Middle Jurassic).

The same problem arises in interpreting these K-Ar results as was encountered in the northern region. Nevertheless, we believe that the ages are recording events from as early as the Middle Jurassic to late Cretaceous, with a fairly well defined event recognized in two widely scattered localities at about 80–85 Ma, possibly related to the last major period of granitic emplacement in the whole region.

B. CONCLUSIONS

The K-Ar ages on plutonic rocks from this region of West Kalimantan indicate a complex history of magmatism and subsequent cooling extending from the Middle Jurassic to the late Cretaceous.

TABLE 1. *Pottassium-argon ages on minerals from mainly granitic intrusive rocks from part of West Kalimantan, south of the Kapuas River, Indonesia*

Field no.	Lab. no.	Material dated	K (wt %)	Rad. ^{40}Ar (10^{-9} mol/g)	100 Rad. ^{40}Ar / Total ^{40}Ar	Calculated age (m.y.) ± 2 s.d.	Rock type	Locality
Schwaner Zone								
K30	73-516	Biotite	7.762,7.739	1.583	95.0	111.4 ± 2.6	Tonalite	
		Hornblende	0.790,0.806	0.1624	89.4	111.0 ± 4.5		
K32	73-518	Biotite	6.891,6.810	1.417	92.7	112.7 ± 2.6	Adamellite	Upstream from Tayan on banks of Kapuas River
		Hornblende	0.443,0.446	0.0943	79.8	115.5 ± 2.6		
K34	73-520	Biotite	7.776,7.760	1.607	95.8	112.8 ± 1.5	Granite	
		Hornblende	0.445,0.452	0.0938	82.0	114.0 ± 3.0		
K35	73-521	Biotite	7.044,7.055	1.351	96.4	104.7 ± 2.4	Tonalite	
K75	73-532	Biotite	6.881,6.910	1.269	95.1	100.6 ± 2.2	Tonalite	Nangah Koman, Sekadau River
K76	73-533	Biotite	7.238,7.254	0.993	86.1	75.5 ± 1.7	Granodiorite	Long Ridge between Kamaru and Kiung Kang Rivers
		Hornblende	0.489,0.477	0.0670	83.2	76.4 ± 3.2		
K74	73-531	Hornblende	0.263,0.257	0.0406	58.5	85.8 ± 3.6	Gabbro	Gunung Biwa
K52	73-1596	Biotite	7.233,7.224	1.172	94.7	89.0 ± 1.3	Tonalite	Keriau River at Nangah Bengaris
		Hornblende	0.502,0.498	0.0870	84.7	95.3 ± 3.0		
Southern Zone								
K14B	73-1595	Biotite	7.092,7.136	1.079	92.2	83.3 ± 1.9	Granodiorite	Islet E of Sempadeh Darat Island
K1	73-506	Biotite	6.599,6.576	1.034	92.2	86.2 ± 2.0	Adamellite	SE side of Cempedak Island
K43	73-523	Biotite	7.392,7.380	2.103	95.6	153.5 ± 3.5	Granite	S of Bukit Belaban Tujuh
K22	73-514	Hornblende	0.821,0.815	0.1915	88.0	127.1 ± 2.8	Granite	Riam, rapids upstream
K15	73-508	Biotite	7.725,7.642	1.119	90.9	80.1 ± 2.0	Granite	
		Hornblende	0.495,0.478	0.0700	73.2	79.1 ± 4.8		
K16	73-509	Biotite	7.775,7.794	1.126	92.0	79.5 ± 1.2	Granite	Riam Kusik area
		Hornblende	0.409,0.403	0.0581	74.5	78.7 ± 3.3		
K17	73-510	Biotite	7.920,7.890	1.167	90.8	81.2 ± 2.0	Monzodiorite	
K19	73-511	Biotite	7.677,7.719	1.135	92.9	81.1 ± 2.0	Granite	

$\lambda_e = 0.585 \times 10^{-10} yr^{-1}$ $\lambda_\beta = 4.72 \times 10^{-10} yr^{-1}$ $^{40}K/K = 1.19 \times 10^{-2}$ atom %

Late Cretaceous radiometric ages have been reported from a number of other localities in the region around West Kalimantan. Granite and amphibolite from Karimata Island have been dated as late Cretaceous (Priem *et al.* 1975). Granitic rocks from West Sarawak, and the Natuna, Tambelan and Anambas islands (Kirk 1968, Haile & Bignell 1971, Haile 1971 and Bignell 1972) have yielded ages mostly in the range 77 to 112 Ma, although there are a few younger ages. There is a concentration of ages around 80 Ma, which Bignell (1972) believes may approximate to the time of emplacement of some of the granites. Further confirmation of this conclusion is provided by the 82 ± 17 Ma Rb-Sr whole rock

TABLE 2. *Palaeomagnetic results from West Kalimantan* $\tilde{B}$*—Magnetic cleaning field. D,I—declination (east of true north) and inclination (positive down) of cleaned remanence.*

Sample No.	Locality	Rock Type	$\tilde{B}$ (mT)	D, I	Polarity	Virtual Pole
K2	110.0E,1.4S	Dyke	30	309, -20	N	8E,38N
K3	110.0E,1.4S	Dyke	30	324, -28	N	347E,52N
K4A	110.0E,1.4S	Metatuff	30	296, +18	N	31E,25N
K4B	110.0E,1.4S	Metatuff	30	98, -18	R	30E, 8N
K6A	110.0E,1.4S	Dyke	30	330, - 4	N	18E,60N
K6B	110.0E,1.4S	Dyke	30	319, + 5	N	25E,49N
K7	110.0E,1.4S	Hornfels	30	93, +17	R	11E, 3N
K9	110.0E,1.4S	Granite	30	348, + 8	N	44E,77N
K10	110.0E,1.4S	Metatuff	60	320, +22	N	39E,48N
K11	110.0E,1.4S	Hornfels	30	310, - 4	N	18E,40N
K12	110.0E,1.4S	Hornfels	30	301, -16	N	11E,31N
K13A	110.1E,1.4S	Dyke	30	36, -61*	N	- -
K13B	110.1E,1.4S	Dyke	30	10, -57*	N	- -
K14A	110.1E,1.6S	Granite	30	300, +10	N	27E,30N
K14B	110.1E,1.6S	Granite	30	301, + 8	N	26E,31N
K23	110.0E,1.3S	Granite	30	316, -29	N	359E,44N
K25	110.0E,1.3S	Dyke	30	288, + 5	N	23E,18N
K26	110.0E,1.3S	Dyke	30	293, +10	N	26E,23N
K27	110.0E,1.3S	Dyke	30	282, +29	N	36E,11N
K28	110.0E,1.3S	Dyke	30	284, +22	N	32E,13N
K29	110.0E,1.3S	Granite	30	306, +34	N	43E,33N
K37	110.5E,1.0S	Tuff	30	175, -19*	R	- -
K38	110.5S,1.0S	Tuff	30	180, -25*	R	- -
K39	110.5E,1.0S	Tuff	30	171, - 8*	R	- -
K40	110.5E,0.9S	Volcanic	30	321, +14	N	33E,50N
K41	110.5E,0.9S	Volcanic	30	331, + 7	N	29E,61N
K47	110.9E,2.3S	Porphyry	30	8, -13*	N	- -
K48	110.9E,2.3S	Porphyry	30	68, -33*	N	- -
K51	105.8E,0.8S	Dyke	20	128, -38	R	43E,35N
K52	105.8E,0.8S	Tonalite	20	340, 0	N	18E,70N
K53	110.8E,10.S	Dyke	20	320, -10	N	14E,50N
K54	110.8E,1.0S	Dyke	20	327, - 9	N	14E,57N
K55	110.8E,1.0S	Dyke	20	311, -20	N	8E,40N
K59	110.7E,1.2S	Dyke	30	284, -25	N	7E,14N
K60	110.7E,1.2S	Dyke	30	289, +12	N	28E,19N
K61	110.6E,1.2S	Dyke	30	312, - 2	N	20E,42N
K62	110.6E,1.2S	Dyke	30	316, +20	N	36E,45N
K63	110.8E,1.2S	Dyke	20	123, -10	R	28E,33N
K64	110.8E,1.2S	Dyke	20	141, +17	R	8E,50N
K66	110.4E,1.5S	Tuff	30	314, +22	N	37E,42N
K67	110.4E,1.5S	Tuff	30	318, +18	N	36E,47N
K71	110.5E,1.4S	Volcanic	30	149, +43	R	340E,52N
K72	110.5E,1.4S	Volcanic	60	131, -29	R	42E,39N
K79	110.3E,1.0S	Tuff	30	140, +40	R	348E,45N
K80	110.3E,1.0S	Tuff	30	155, +28	R	350E,62N
K81	110.3E,1.0S	Tuff	30	168, +44	R	314E,63N

* Measurements omitted in final calculation (table 3).

isochron on the Bukit Tabong-Tinteng Bedil granite in West Sarawak (Bignell 1972).

In the Malay Peninsula a granite from Malacca gave a Rb-Sr isochron age of 79 ± 2 Ma and a K-Ar biotite age of 70 ± 5 Ma, and other similar ages have been reported from the Peninsula (Bignell 1972, Bignell & Snelling, in press). From wells in the Java Sea north of Java, Katili (1973) has recorded evidence of an event around 100 Ma ago, and Rb/Sr (biotite) ages of around 88 Ma for granite from Lampong, South Sumatra.

3. Palaeomagnetic results

Of the forty eight oriented samples collected for palaeomagnetic studies only two failed to give satisfactory data. All of these forty six samples were extremely stable under progressive alternating field demagnetization and standard cleaning procedures were applied in either 20 or 30 mT. The great majority of samples (thirty nine out of forty six) gave directions of magnetization after cleaning along a northwest—southeast axis (Fig. 4 and Table 2). The seven remaining samples have directions that tend to lie along a north-south axis, suggesting they may be younger, possibly Tertiary. The main group probably represents the direction of magnetization acquired when these rocks originally cooled during the Cretaceous, since two of this group are among those dated radiometrically as late Cretaceous, and six others are hornfels and metatuff in the contact zone of late Cretaceous adamellite. Apart from the hornfels and metatuff a wide variety of rock types are represented in the main group including granites, basic dykes, volcanics and tuffs.

The dykes which cut the granite are younger than the granite by an unknown amount and there is no direct evidence for the relative age of the other samples. If the samples are subdivided into groups consisting of basic rocks (dykes, tuffs and volcanics) and acid rocks (granites and associated hornfels and metatuffs) there is no difference between their mean directions (Table 2). From this it is concluded that the acid and basic rocks were emplaced during the same igneous cycle and that all the samples are probably about the same age, namely late Cretaceous. The overall mean of the thirty nine sample directions is therefore

TABLE 3. *Overall mean palaeomagnetic results from the igneous rocks of West Kalimantan. R-length of resultant of N unit vectors; k-precision parameter;* α_{95}*-radius of circle of* 95 *percent confidence about mean*

Combination	N	D	I	R	k	α_{95}	Palaeomagnetic Pole
Basic rocks	28	314	-2	24.766	8.3	10.0	20E,44N
Acid rocks	11	304	+5	9.867	8.8	16.2	24E,34N
All directions	39	311	0	34.490	8.4	8.3	21E,41N*
All VGPs	39	-	-	36.288	14.0	6.3	21E,41N

* Assumes mean site position 110.0E,1.2S. Polar error is dp = 4.2°, dm = 8.3°

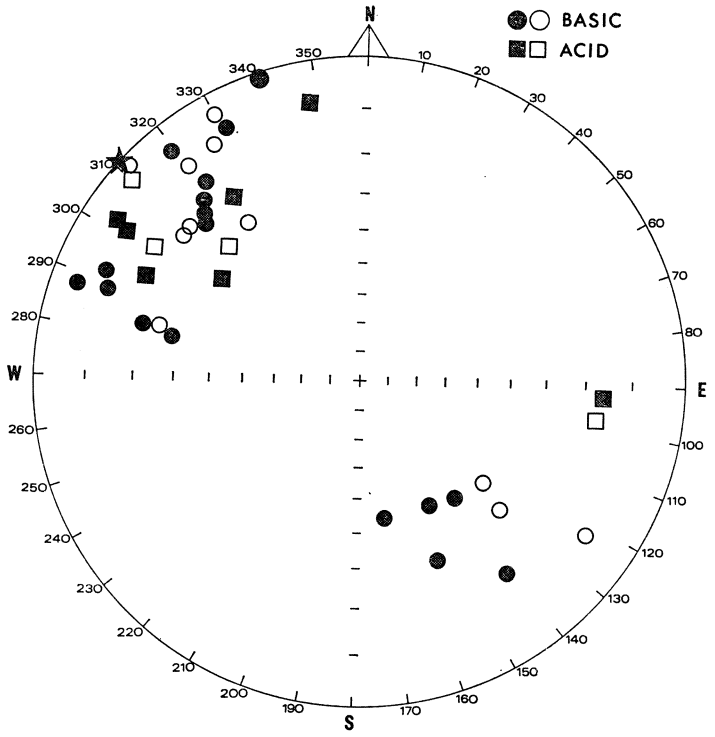


FIG. 4. Stereographic projection of directions of magnetization observed in the main group of thirty nine samples. Solid symbols are on the lower hemisphere and open symbols on the upper hemisphere.

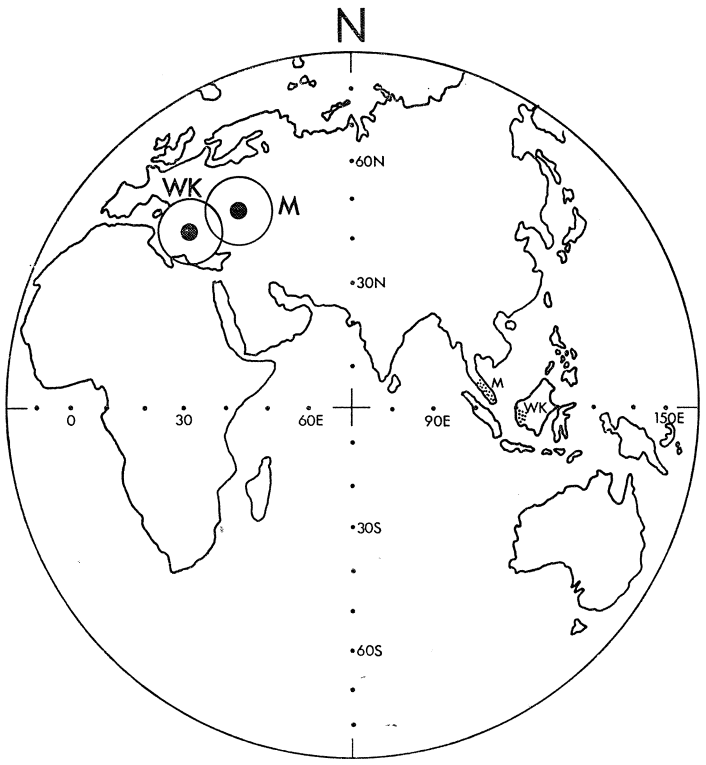


FIG. 5. Palaeomagnetic pole positions derived from Cretaceous rocks of the Malay Peninsula (M) and West Kalimantan (WK). Circles of 95% confidence are shown.

considered to be the best estimate of the palaeomagnetic field in West Kalimantan during the late Cretaceous (Table 3).

The mean Cretaceous palaeolatitude is 0° for West Kalimantan (Table 3) and 17° for the southern part of the Malay Peninsula (McElhinny, Haile & Crawford 1974) as shown in Fig. 1. Using the polar errors of 6·1° and 8·7° assigned to the pole from each region respectively, this gives a palaeolatitude difference of 17° ± 10·4° between the two sampling regions. This is not signicantly different from the present angular difference between the two regions of 8·0°.

The overall palaeomagnetic pole position deduced from these Cretaceous igneous rocks of West Kalimantan is plotted on the present latitude-longitude grid in Fig. 5. For comparison the pole position derived from Cretaceous lavas and dykes of the Malay Peninsula (McElhinny, Haile & Crawford 1974) is also shown. The circles of confidence overlap and, bearing in mind that the error assigned to the present result is probably too low, there is no significant difference between them. The results indicate that the two regions are now sited in about the same position relative to each other, and occupy much the same latitude as in the Cretaceous, but have together rotated anticlockwise 45 to 50°. Probably West Kalimantan and the Malay Peninsula have been parts of the same structural unit since the Middle Cretaceous.

In marked contrast Australia lay much further south and was attached to Antarctica in the Cretaceous. During the Tertiary Australia drifted northwards experiencing a latitude change of 35° since the Middle Cretaceous (McElhinny, Embleton & Wellman 1974). The drift history has therefore been one of convergence between Australia and South-east Asia during the Tertiary.

4. Conclusions

The investigation indicates that an important event, probably of intrusion of granitic to gabbroic rocks, occurred in the late Cretaceous in both the Schwaner Zone and Southern Region, centered around 80–85 Ma ago. This late Cretaceous event is also recognized regionally.

Palaeomagnetic measurements indicate that West Kalimantan and the Malay Peninsula have behaved as a unit since the Middle Cretaceous, have remained at about their present latitudes, but have rotated anti-clockwise about 50° since then.

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R. Rudowski carried out the mineral separations and Robyn Maier and Zarko Roksandic assisted in the determination of the K-Ar ages. David Edwards assisted with the magnetic measurements.

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